Surface Areas and Volumes — MCQ Worksheet

Mathematics • Chapter 13 • *25 Questions, 1 mark each — Total: 25 marks*

Instructions: *Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.*

Q1. Volume of a cube of side a is:

(A) $6a^2$ (B) a^2
(C) a^3 (D) $4a^2$

Q2. Total surface area of a cube of side a is:

(A) 6a (B) $6a^2$
(C) a^2 (D) $4a^2$

Q3. Volume of a cuboid with dimensions l, b, h is:

(A) lbh (B) 2(lb + bh + hl)
(C) l + b + h (D) l + bh

Q4. Total surface area of a cuboid is:

(A) 2(lb + bh + hl) (B) lbh
(C) 6lb (D) 2lb

Q5. Volume of a cylinder of radius r and height h is:

(A) $\pi r^2 h$ (B) $2\pi rh$
(C) $\pi r^2 + h$ (D) $2\pi r + h$

Q6. Curved surface area of a cylinder is:

(A) $2\pi rh$ (B) $\pi r^2 h$
(C) $2\pi r^2$ (D) πrh

Q7. Total surface area of a closed cylinder is:

(A) $2\pi r(r + h)$ (B) $2\pi rh$
(C) $\pi r^2 h$ (D) πrh

Q8. Volume of a sphere of radius r is:

(A) $4/3\ \pi r^3$ (B) $4\pi r^2$
(C) πr^2 (D) πr^3

Q9. Surface area of a sphere of radius r is:

(A) $4\pi r^2$ (B) $4/3\ \pi r^3$
(C) $2\pi r^2$ (D) πr^2

Q10. Surface area of a hemisphere (curved only) is:

(A) $2\pi r^2$ (B) $4\pi r^2$
(C) πr^2 (D) $3\pi r^2$

Q11. Total surface area of a closed hemisphere is:

(A) $2\pi r^2$ (B) $3\pi r^2$
(C) $4\pi r^2$ (D) πr^2

Q12. Volume of a hemisphere is:

(A) $2/3\ \pi r^3$ (B) $4/3\ \pi r^3$
(C) πr^3 (D) $2\pi r^3$

Q13. Volume of a cone of radius r and height h is:

(A) $1/3\ \pi r^2 h$ (B) $\pi r^2 h$
(C) $1/2\ \pi r^2 h$ (D) $2\pi r^2 h$

Q14. Curved surface area of a cone (slant height l) is:

(A) πrl (B) $\pi r^2 l$
(C) $2\pi rl$ (D) πrh

Q15. Total surface area of a cone is:

(A) $\pi r(r + l)$ (B) πrh
(C) $\pi r^2 + 2\pi rl$ (D) $\pi r^2 + \pi rl$

Q16. Slant height of a cone with radius r, height h is:

(A) $\sqrt{}(r^2 + h^2)$ (B) $r + h$
(C) $r^2 + h^2$ (D) $\sqrt{}(r^2 - h^2)$

Q17. A cone, hemisphere and cylinder stand on equal bases and have equal heights. Ratio of their volumes is:

(A) 1:2:3 (B) 1:1:1
(C) 2:3:1 (D) 3:2:1

Q18. If a cylinder and cone have same base and height, ratio of their volumes is:

(A) 1:1 (B) 3:1
(C) 1:3 (D) 2:1

Q19. A solid of dimension 4 × 3 × 2 cm is melted into a cube. Side of cube is:

(A) $\sqrt{}24$ cm (B) ■24 cm
(C) 6 cm (D) 12 cm

Q20. A right circular cylinder has radius 7 cm and height 10 cm. Its volume ($\pi = 22/7$) is:

(A) 1540 cm^3 (B) 154 cm^3
(C) 440 cm^3 (D) 2200 cm^3

Q21. A cylinder of radius 5 cm and height 14 cm. Curved surface area ($\pi = 22/7$):

(A) 440 cm^2 (B) 220 cm^2
(C) 154 cm^2 (D) 880 cm^2

Q22. Volume of a sphere with radius 7 cm ($\pi = 22/7$) is:

(A) 1437.33 cm^3 (B) 4188 cm^3
(C) 1372 cm^3 (D) 1078 cm^3

Q23. A toy is in the shape of a cone surmounted on a hemisphere. Its total volume is:

(A) $1/3\ \pi r^2h + 2/3\ \pi r^3$ (B) $\pi r^2h + \pi r^2h$
(C) $\pi r^2h + \pi r^2$ (D) $2\pi rh + \pi r^2$

Q24. If a cone has its height doubled keeping the radius same, its volume:

(A) Stays the same (B) Doubles
(C) Triples (D) Halves

Q25. If radius of a sphere is doubled, its surface area becomes:

(A) 2 times (B) 3 times
(C) 4 times (D) 8 times

— End of Worksheet —